Confidence level, not metacognitive sensitivity, is the domain-general component of confidence: a measurement-lens re-analysis [PLACEHOLDER TITLE]

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1 Abstract

Note

[STUB — written in dispatch 012 after Ben's review.]

2 Introduction

Note

[STUB — written in dispatch 012.]

3 Methods

3.1 Primary dataset and provenance

The primary data are the trial-level records of A. E. Lund, C. M. C. Corrêa, F. Fardo, S. M. Fleming, and M. G. Allen [1], a confirmatory multi-domain metacognition study spanning four domains — episodic memory, visual perception, and two semantic-knowledge judgements (Calories and GDP). We re-fetched the trial file directly from the authors' public repository at a pinned commit (0c38aba4eb…) and verified it before any use: the MD5 checksum matched the value recorded by the authors (500b2e6e1eb0e47e717feba5b912ad9f), and the structure matched expectation — 193,200 trials from 322 subjects across the four domains, with confidence reported on a 1-7 scale. As an external cross-check (never as a primary number) we reproduced per-subject mean confidence against the authors' shipped MLE fits; agreement was high (within-domain correlations ≥ 0.998), the small residual reflecting their undocumented preprocessing. Median trials per subject are 200 for memory and vision and roughly 100 for the two knowledge domains. Following the from-scratch mandate of this project, all analysis code is original to this repository and no computational result from any prior effort was reused or consulted; every reported number is computed in code from the verified raw data.

3.2 The two components and their measures

We operationalize confidence LEVEL (set-point) as the subject's scale-normalized mean confidence within a domain, and metacognitive SENSITIVITY (informativeness) as AUROC2 — the area under the type-2 ROC [2], [3], a model-light index of how well confidence discriminates a person's own correct from incorrect

responses. We treat the signal-detection meta-d′/M-ratio family [4] as secondary rather than the spine of the paper, consistent with the model-light posture of this project. Because AUROC2 is confounded by first-order performance (d′), **every sensitivity claim in this paper is d′-partialed**: sensitivity is the residual of AUROC2 after regressing out same-domain d′. This choice is not cosmetic — it was forced by the Step 0 gate (below), where the raw (un-partialed) latent estimator converted a pure d′ correlation into a spurious sensitivity correlation. We do not use gamma as a sensitivity measure: in recovery simulation it leaked the set-point (a level-to-sensitivity contamination), so it was dropped despite its long use as a metacognitive index [5]. The latent cross-domain correlations are estimated with an exchangeable-parcel confirmatory factor model [6] that splits each subject-domain's trials into parcels, giving a measurement model with a built-in reliability correction rather than a single noisy point measure.

3.3 Simulation-gate program (Step 0)

Before testing the thesis we required a foundation gate: prove the estimators recover known planted values, and prove the two components are separable (estimating one does not leak into the other). The gate was run in rounds and we report its failures honestly. The first round established that level is recovered near-perfectly (recovery r ≈ 0.99, no leakage) but that AUROC2 recovery is reliability-capped at realistic trial counts and that the **raw** level-vs-sensitivity contrast manufactures a gap even under a true null — so the naive contrast is invalid as designed. The second round showed that a d′-partialed exchangeable-parcel latent estimator passes all criteria at the 200/200-trial configuration but that raw-latent variants convert the d′ confound into spurious sensitivity correlation — establishing d′-partialing as mandatory. The final round calibrated the corrected estimator on training seeds and evaluated it blind on fresh seeds: at 200 trials per task in both domains the estimator passed every criterion, so the memory × vision pair (the only pair with 200/200 trials) is the sole **confirmatory** pair; other pairs are descriptive with explicit bias bands. The calibration coefficients were then **frozen** (intercept a = 0.0401, slope b = 0.7492) and never refit. A power note carried forward from the gate: at a true gap of 0.4 and the 200/200 configuration, the corrected test excludes zero in about 60% of simulated replicates — the confirmatory test is powered to detect a substantial gap but not a small one.

3.4 Confirmatory design

The single preregistered confirmatory test is the memory × vision pair, chosen mechanically as the only pair meeting the gate's 200/200-trial requirement, and fixed by git timestamp before the cross-domain quantities were computed. We estimate the latent cross-domain correlation of level (ρ_level) and of d′-partialed sensitivity (ρ_sens) in the parcel-CFA framework, apply the frozen calibration correction to ρ_sens, and form the primary outcome — the gap ρ_level − ρ_sens — with a delta-method confidence interval; a subject bootstrap (B = 2000, calibration held fixed) is the secondary interval. A firewall held until this step: no cross-domain quantity of any kind was computed while the measurement layer and reliabilities were being built. Wording is fixed in advance: a gap whose interval includes zero is reported as **not excluded**, never as absent.

3.5 Descriptive tier

The other five domain pairs did not meet the 200/200 gate and are therefore **descriptive, non-confirmatory**. For each we report ρ_level and the **uncorrected** ρ_sens (no valid calibration exists at their trial configurations), and we attach a frozen systematic-bias band to every gap — ±0.09 for pairs at 200/100 trials and ±0.10 for the Calories × GDP (100/100) pair — carried directly from the gate's measured bias at those configurations. These bands are systematic-bias allowances, not confidence intervals; a descriptive gap is treated as consistent with zero unless it clears both its sampling CI and its band.

3.6 Non-ipsative multilevel decomposition (Step 4)

The classic trap in this literature is the ipsative artifact: subtracting each person's mean confidence mechanically induces negative between-domain correlations, which can masquerade as a domain-specific bias. We therefore do **not** test the thesis by mean subtraction. Instead we fit a trial-level multilevel model [7] that simultaneously estimates a person-general set-point, a person×domain set-point residual, a person-general confidence-resolution slope (Δ confidence for correct vs. incorrect responses, domain-accuracy centered), and a person×domain resolution residual. From the variance components we form two generalizability indices — ICC_level and ICC_sens, the person-general share of set-point and of resolution variance — with cluster-bootstrap intervals (B = 500). This is a variance-share estimand, distinct from the Step 3 latent correlation gap and not a re-test of it. Robustness rungs re-fit with within-domain z-scored confidence (R1) and an ordinal cumulative-link mixed model (R2). As a methods warning we also compute what the discouraged naive mean-subtraction analysis would have produced, on the same trials and in matched simulation, to exhibit the artifact the model is built to avoid. The resolution slope is not d′-partialed; the residual d′ confound is quantified in diagnostics rather than removed, and is flagged wherever the slope is discussed.

3.7 Confidence Database: study selection

For an external robustness tier we used the Confidence Database [8], fetched from OSF and verified file-by-file (255,175,101 bytes across all files; 0 MD5 mismatches). Provenance is drawn from the committed manifest and provenance JSON, not from any prior narrative: the live OSF node has no registered snapshot, so it is pinned by access date and per-file version, and 8 files carry OSF versions above 1 (all four selected studies' data CSVs are version 1). Study selection was mechanical and pre-specified: of 180 parsed study files, 21 have a within-subject multi-task structure, and a four-criterion funnel (a task pair with enough qualifying subjects and trials; a usable confidence scale; both accuracy classes present in every task; no structural disqualifier) selects 4 studies: Arbuzova_2021_expt2 [9], Arbuzova_unpub_3, Bang_2019_Exp2 [10], and Haddara_2022_Expt1 [11]. A firewall again held: no cross-task quantity was computed during selection or measurement.

3.8 Confidence Database: contrast and yardsticks

Taxonomy (binding). This tier tests TASK-generalization and contains exactly one external cross-**domain** pair. Only Arbuzova_unpub_3 memory × visual is cross-domain. Arbuzova_2021_expt2 is a visuomotor condition × condition pair; Bang_2019_Exp2 (coarse vs. fine Gabor) and Haddara_2022_Expt1 (letter vs. color grids) are cross-**task** pairs within vision. These four studies are never presented as four cross-domain replications.

For each study we correlate confidence level across the task pair and d′-partialed AUROC2 across the same pair (subjects with ≥ 60 clean trials in both tasks, listwise complete; subject bootstrap B = 2000; residualization recomputed inside every resample). Because no calibration correction exists for these configurations, nothing in this tier is confirmatory. Two attenuation-aware yardsticks replace the missing correction: (i) a disattenuation **range** for each correlation (r divided by the geometric mean of the two reliabilities, evaluated at the reliability point estimate and both CI bounds — a sensitivity range, never a corrected point value); and (ii) a per-study **null-reference interval** — the 2.5–97.5% span of the raw gap in a simulated true-null world (true ρ_level = true ρ_sens) matched to each study's n, trials, confidence scale, and reliabilities, quantifying how much raw-gap inflation differential attenuation alone produces. The preregistered reading rule: the Lund pattern **holds** in a study only if r_level exceeds the top of r_sens's disattenuation range **and** the raw gap exceeds the null-reference interval under both null scenarios; one condition met is **partial**; neither is **fails**. No meta-analytic pooling is performed across the four heterogeneous studies. Bang's measures were

recomputed from raw with catch trials dropped; Haddara's between-subject feedback groups were derived from raw for a heterogeneity check only.

4 Results

4.1 The measurement floor

Across all four Lund domains, confidence level is measured far more reliably than metacognitive sensitivity (Table 1). Split-half Spearman–Brown reliability [12], [13] of level ranges 0.95-0.99, while AUROC2 reliability ranges only 0.39-0.84. The gap is even starker in trials-needed terms: reaching a reliability of 0.7 takes on the order of 7–15 trials for level but roughly 106 (memory) to 362 (vision) trials for AUROC2. At the parcel level used by the latent model, AUROC2 reliabilities are lower still (memory 0.55, vision 0.27, Calories 0.31, GDP 0.24). This asymmetry is the engine of every inflated raw gap that follows: a difference between two correlations is biased upward when one variable is measured much more reliably than the other.

Table 1: Split-half (Spearman–Brown) reliability of confidence level vs. metacognitive sensitivity (AUROC2), per Lund domain. Sensitivity is measured much less reliably than level in every domain.

	Domain	Level (mean conf.)	Sensitivity (AUROC2)
memory	Memory	0.97	0.84
vision	Vision	0.99	0.59
Calories	Calories	0.95	0.47
GDP	GDP	0.95	0.39

4.2 Confirmatory test: memory × vision

CONFIRMATORY. In the single preregistered pair, confidence level generalizes strongly across domains (ρ_level = 0.42, 95% CI [0.33, 0.51]). Corrected metacognitive sensitivity also correlates across domains and its interval excludes zero (ρ_sens = 0.31, 95% CI [0.08, 0.54]) — sensitivity is not domain-specific either. The primary outcome, the level-minus-sensitivity gap, is 0.11 with 95% CI [−0.14, 0.36] (n = 317): **the null gap is not excluded**. Given roughly 60% power at a true gap of 0.4, this is a "not excluded," not an "absent." The subject bootstrap agrees (gap CI [−0.12, 0.38]).

The corrected result is what keeps this honest. The uncorrected latent sensitivity correlation is 0.27 (95% CI [0.10, 0.44]), giving an uncorrected gap of 0.15; and the fully naive subject-level analysis — raw mean-confidence correlation 0.40 minus raw AUROC2 correlation 0.20 — yields a raw gap of 0.20 with 95% CI [0.05, 0.35] (n = 320) that **excludes zero**. That naive analysis would have affirmed the thesis; the correction machinery, validated in simulation before it touched these data, is exactly what prevented the claim (Table 5, row `naive_gap_exhibit`).

4.3 Descriptive landscape

DESCRIPTIVE (NON-CONFIRMATORY). Across the five non-confirmatory pairs (Table 2, Figure 1), level generalization is uniform (ρ_level range 0.32-0.47) while the gap is heterogeneous and vision-anchored: positive where vision is involved, near zero or negative for the memory × knowledge pairs. No descriptive gap clears both its sampling CI and its systematic-bias band jointly. The most positive gap, vision × GDP (0.30), has a CI that excludes zero, but its lower bound (0.06) lies inside its ±0.09 band, so no nonzero-gap claim is made for it or any descriptive pair. Read through the measurement lens (Figure 2, exploratory), the level-advantage appears where sensitivity is least reliably measured and reverses where it is best measured (the memory pairs) — one measurement fact, not six psychological ones.

Table 2: Descriptive tier (NON-CONFIRMATORY): five domain pairs. rho_sens is uncorrected; each gap carries a frozen systematic-bias band. No gap clears its CI and band jointly.

Pair	rho_level [95% CI]	rho_sens (uncorr.) [95% CI]	gap [95% CI]	band
memory x Calories	0.38 [0.28, 0.47]	0.40 [0.20, 0.59]	−0.02 [−0.24, 0.20]	+/-0.09
memory x GDP	0.32 [0.22, 0.43]	0.46 [0.24, 0.67]	−0.13 [−0.37, 0.11]	+/-0.09
vision x Calories	0.45 [0.36, 0.54]	0.26 [0.05, 0.47]	0.19 [−0.03, 0.42]	+/-0.09
vision x GDP	0.46 [0.37, 0.55]	0.17 [−0.06, 0.39]	0.30 [0.06, 0.54]	+/-0.09
Calories x GDP	0.47 [0.38, 0.56]	0.24 [−0.02, 0.50]	0.24 [−0.04, 0.51]	+/-0.10

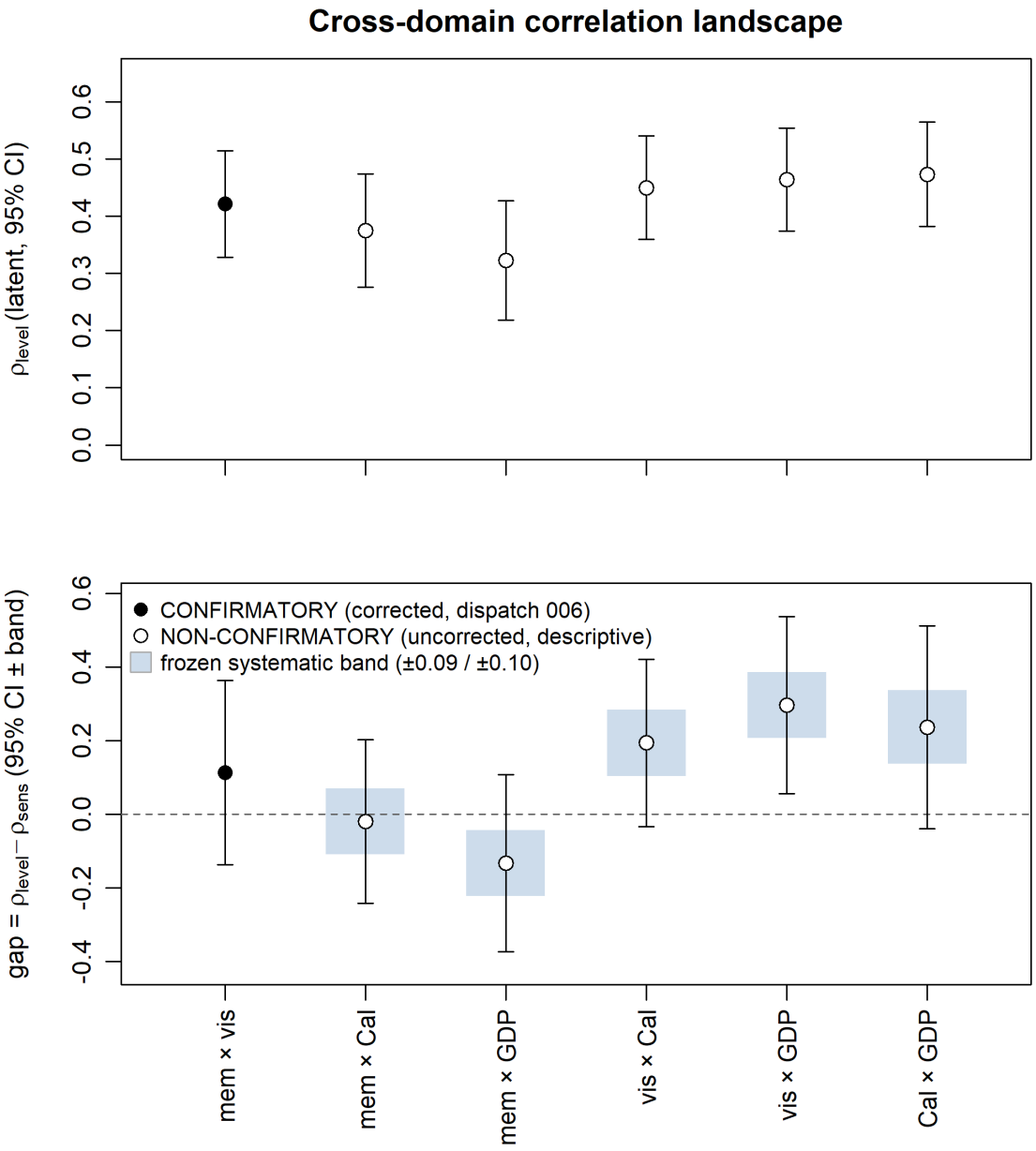


Figure 1: Cross-domain correlation landscape (NON-CONFIRMATORY except the filled memory x vision confirmatory point). Full caption: figures/fig_captions.md (Figure 1).

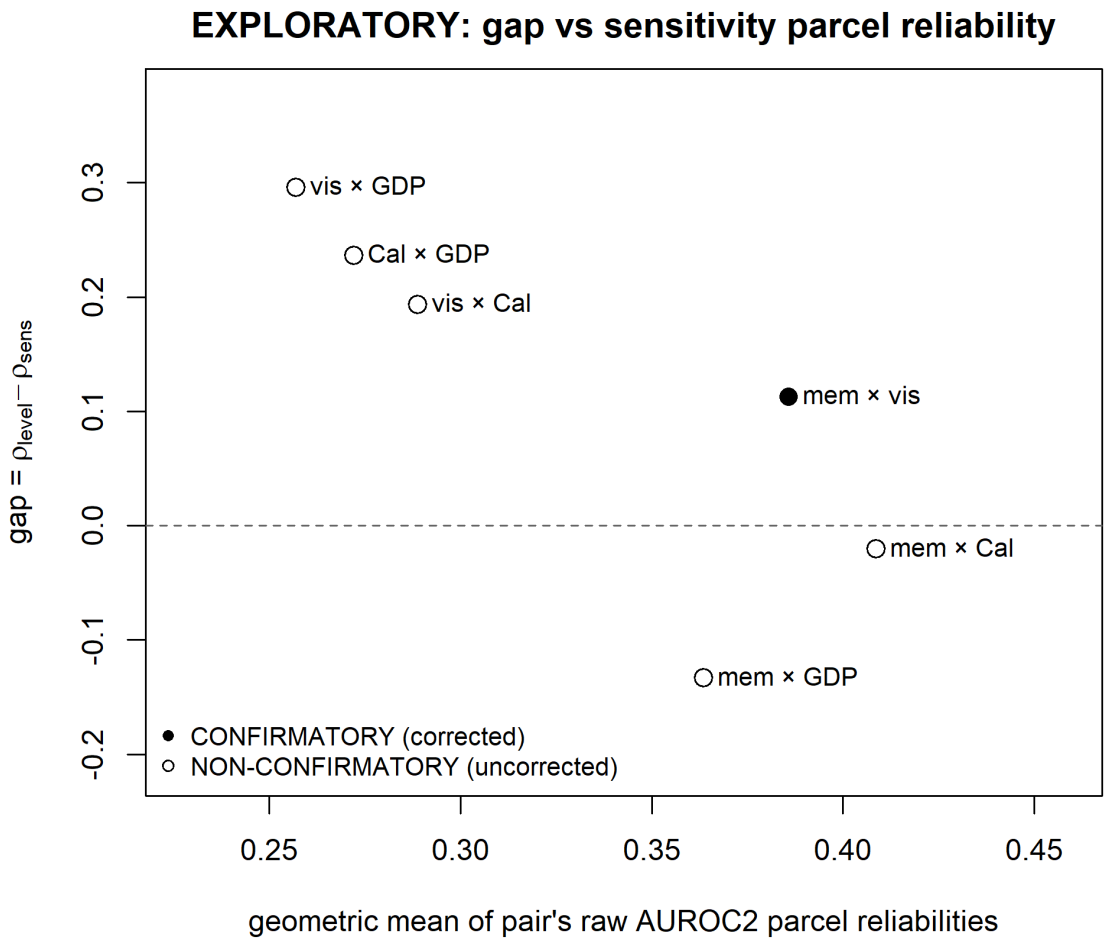


Figure 2: EXPLORATORY: gap vs. sensitivity measurement quality. No line fitted, no test performed. Full caption: figures/fig_captions.md (Figure S1).

4.4 Multilevel decomposition

STEP 4 MULTILEVEL (NON-CONFIRMATORY). The non-ipsative model apportions confidence variance into person-general and person×domain shares. The person-general share is 0.44 for the set-point (ICC_level) and 0.34 for resolution (ICC_sens); their difference is Δ_ICC = 0.11 with 95% cluster-bootstrap CI [0.001, 0.210]. A Δ_ICC of zero sits at the very edge of this interval — we present this as a boundary result and never claim the zero-exclusion; it is directionally consistent with the confirmatory corrected gap (two different estimands, the same "cannot quite exclude zero" verdict). Robustness rungs agree (R1 z-scored: ICC_level 0.44, ICC_sens 0.36; R2 ordinal: 0.43, 0.32). The methods exhibit confirms why mean subtraction was avoided: the naive grand-mean-centered correlations run from −0.67 to 0.08 (mean −0.30) in these data, and in matched simulation the same computation manufactures a mean of about −0.31 under a planted null (mechanical expectation ≈ −1/3), while the multilevel model recovers the planted shares.

4.5 External robustness: the Confidence Database

CONFDB ROBUSTNESS (NON-CONFIRMATORY). This tier tests task-generalization with exactly one external cross-domain pair; the within-vision task pairs and the condition × condition pair are not cross-domain replications. Confidence level travels well: r_level is uniformly positive and precisely estimated

across the four studies (0.55–0.87; Table 3, Figure 3). d′-partialed sensitivity is heterogeneous. Because level is measured near-perfectly (split-half 0.97-1.00) and AUROC2 is not (0.62-0.90), a positive raw gap is expected even under a true null, so each raw gap must be read against its simulated null-reference interval (Table 4): at these configurations, differential attenuation plus sampling noise alone can produce raw gaps up to ~0.5 in the small-n studies.

By the preregistered rule, the Lund pattern **holds in 1 of 4** studies, is **partial in 3**, and **fails in 0**. In every study r_level exceeds the top of r_sens's disattenuation range; what separates "holds" from "partial" is whether the raw gap also clears its null reference. Only the single external cross-domain pair does: Arbuzova_unpub_3 memory × visual (r_level = 0.56, 95% CI [0.25, 0.76]; r_sens = −0.10) has a raw gap of 0.551 that exceeds its null-reference upper bounds (0.505 and 0.531) — but only narrowly, by margins of 0.046 and 0.020 against the ρ = 0.30 and ρ = 0.45 references. This single cross-domain pair is therefore consistent with the Lund direction and exceeds its simulated true-null reference, narrowly; a null is not excluded in the three partial studies. Notably, Haddara's within-vision sensitivity correlation is itself high (r_sens = 0.56): metacognitive sensitivity can generalize strongly across near-identical tasks within a domain, which is task generalization, not cross-domain generality.

Table 3: Confidence Database contrast (NON-CONFIRMATORY, robustness). Subject bootstrap B = 2000. Only Arbuzova_unpub_3 memory x visual is cross-domain.

Study	Pair	n	r_level [95% CI]	r_sens (d'-part.) [95% CI]	verdict
Arbuzova_2021_exp1	task1 x task2 (visuomotor, condition x condition)	41	0.55 [0.22, 0.80]	0.24 [−0.08, 0.51]	partial
Arbuzova_unpub_3	memory x visual (CROSS-DOMAIN)	40	0.56 [0.25, 0.76]	−0.10 [−0.35, 0.14]	holds (narrowly)
Bang_2019_Exp2	task1 x task2 (coarse x fine Gabor, within vision)	201	0.57 [0.45, 0.68]	0.07 [−0.05, 0.19]	partial
Haddara_2022_Exp1	task1 x task2 (letter x color grids, within vision)	443	0.87 [0.84, 0.90]	0.56 [0.49, 0.63]	partial

Table 4: Attenuation-aware yardsticks: disattenuation ranges (sensitivity ranges, not corrected point values) and SIMULATION null-reference intervals for the raw gap. Only Arbuzova_unpub_3 exceeds its null-reference interval.

Study	r_level disatt. range	r_sens disatt. range	raw gap [95% CI]	null-ref (rho=.30)	null-ref (rho=.45)
Arbuzova_2021_exp2	[0.53, 0.55]	[0.26, 0.31]	0.14 [−0.22, 0.47]	[−0.36, 0.52]	[−0.31, 0.47]
Arbuzova_unpub_3	[0.57, 0.59]	[−0.25, −0.12]	0.55 [0.19, 0.87]	[−0.32, 0.50]	[−0.24, 0.53]
Bang_2019_Exp2	[0.58, 0.59]	[0.08, 0.09]	0.17 [0.02, 0.32]	[−0.12, 0.25]	[−0.06, 0.26]
Haddara_2022_Exp1	[0.88, 0.88]	[0.64, 0.69]	0.13 [0.09, 0.17]	[−0.08, 0.18]	[−0.04, 0.19]

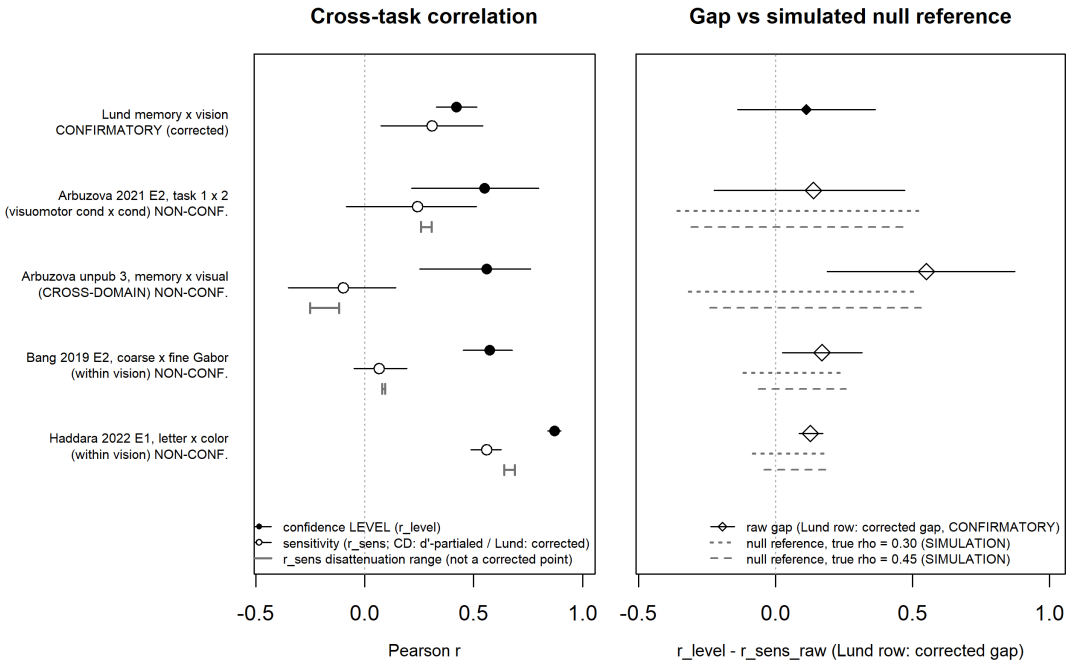


Figure 3: Confidence Database level-vs-sensitivity contrast with the Lund confirmatory pair for comparison (NON-CONFIRMATORY). Full caption: figures/fig_captions.md (Figure 2).

4.6 Claim-status summary

Table 5 collects every quantitative claim with its tier, estimate, and frozen verdict wording (full machine-generated table with sources and CIs: `results/step17_claim_status.md`).

Table 5: Claim-status summary (generated by analysis/17_claim_status.R from committed CSVs). Every value is assert-guarded in code; verdict wording is preregistered.

#	Claim	Tier	Estimate	Verdict (frozen)
1	gap_confirmatory	CONFIRMATORY	0.113 [−0.138, 0.364]	null gap not excluded
2	rho_level_confirmatory	CONFIRMATORY	0.422 [0.328, 0.515]	confidence level is domain-general
3	rho_sens_confirmatory	CONFIRMATORY	0.308 [0.075, 0.542]	corrected sensitivity correlation excludes zero - sensitivity is not domain-specific either
4	naive_gap_exhibit	METHODS EXHIBIT	0.2 [0.051, 0.353]	the uncorrected analysis would have affirmed the thesis; the correction machinery prevented that claim
5	descriptive_landscape	DESCRIPTIVE (NON-CONFIRMATORY)	–	heterogeneous, vision-anchored; no nonzero-gap claim for any descriptive pair
6	step4_decomposition	STEP4 MULTI-LEVEL (NON-CONFIRMATORY)	0.105 [0.001, 0.21]	a Delta_ICC of zero sits at the very edge of the interval; directionally consistent with the confirmatory corrected gap; the zero-exclusion is never claimed
7	ipsative_exhibit	METHODS EXHIBIT	−0.667	grand-mean centering manufactures negative within-subject correlations; asymmetric under unequal trial counts
8	confdb_tier	CONFDB ROBUSTNESS (NON-CONFIRMATORY)	–	holds 1/4, partial 3/4, fails 0/4; the only external cross-domain pair is Arbuzova_unpub_3 memory x visual (consistent with the Lund direction, exceeds its simulated true-null reference narrowly); within-vision task pairs and the condition x condition pair are NOT cross-domain replications
9	reliability_floor	METHODS (NON-CONFIRMATORY)	–	the measurement asymmetry that inflates raw gaps is universal in these data

5 Discussion

Note

[STUB — written in dispatch 012.]

6 Limitations

Note

[STUB — prose written in dispatch 012. Standing checklist seeded below.]

7 Data and code availability

All analysis code is original to this project and public at `https://github.com/dondon20201/paper2`; the analysis runs end-to-end under a renv lockfile (R 4.6.1) [14]. Every reported number is written to a file under `results/` with a name traceable from the text, and each figure's underlying data is saved alongside it (`results/fig_landscape_data.csv`, `results/figS_gap_vs_reliability_data.csv`, `results/fig_confdb_contrast_data.csv`).

Primary data. A. E. Lund, C. M. C. Corrêa, F. Fardo, S. M. Fleming, and M. G. Allen [1], file `data_summary/metacognition_trialData_master.csv` from the repository `embodied-computation-group/dg-metacognition`, pinned at commit 0c38aba4eb… and verified by MD5 (500b2e6e1eb0e47e717feba5b912ad9f). The raw file is not redistributed here (provenance in `data/raw/lund2025/README.md`).

Secondary data. The Confidence Database [8] from OSF node `s46pr`, verified file-by-file against OSF MD5s (255,175,101 bytes, 0 mismatches); the live node has no registered snapshot, so it is pinned by access date and per-file OSF version in `results/step11_CONFDB_manifest.csv`. Raw Confidence Database files (~255 MB) are not redistributed here.

8 References

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